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Klasse: 6ACIF		Übung Nr.: 4_2
Lehrer: Prof. Spanring & Horny	Abgabedatum: 26.04.2025	Raum Nr.: Platz Nr.: Bewertung:



Laborübung Layer-3-Switch am Gerät (Teil 1)

1) Aufgabenstellung

- 4 PCs, verteilt auf 2 VLANs
- Ein Layer 3 Switch verwaltet VLANs und Routing
- VLAN 10: Marketing (192.168.10.0/24), Switchport 1 und 2
- VLAN 20: Technik (192.168.20.0/24), Switchport 3 und 4
- Alle PCs sollen ihre IP-Adresse automatisch per DHCP vom Layer 3 Switch erhalten
- PC1 (VLAN 10) soll PC3 (VLAN 20) pingen können → Routing muss funktionieren
- Keine externe Routerverbindung notwendig

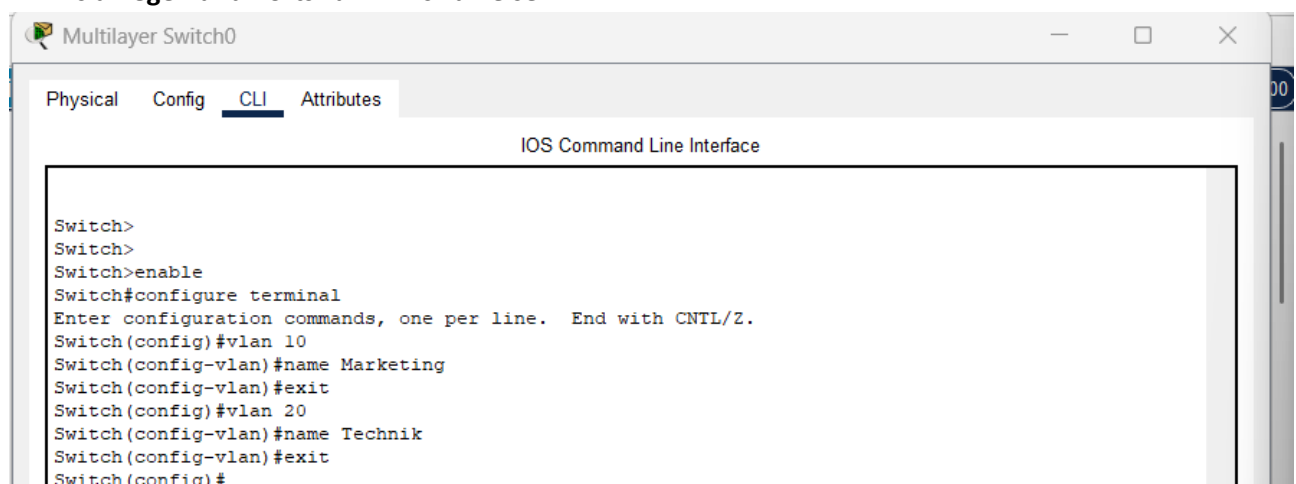
Hinweis:

Für die IP-Adressierung wurde meine KN-Nummer 8 verwendet.

Daher sind die verwendeten Netze 192.168.8.0/24 und 192.168.80.0/24.

2) Switch-Konfiguration

VLANs anlegen und Ports zu VLANs zuweisen:



```
Switch>
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Technik
Switch(config-vlan)#exit
Switch(config)#
```

```
Switch(config)#interface range GigabitEthernet1/0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range GigabitEthernet1/0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

SVI konfigurieren:

```
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.8.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
```

```
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.80.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
```

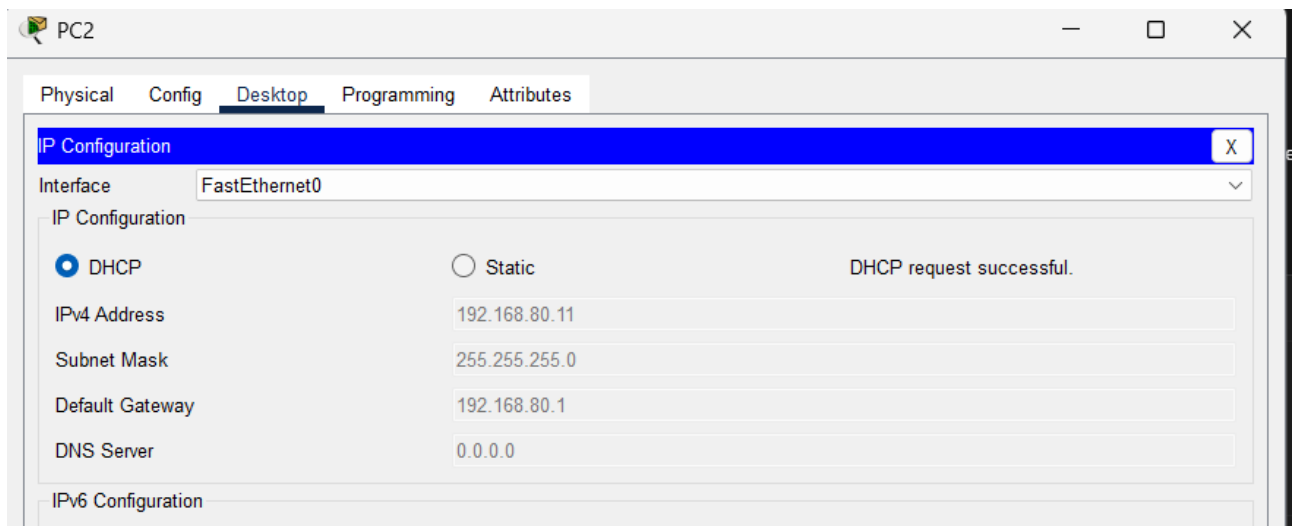
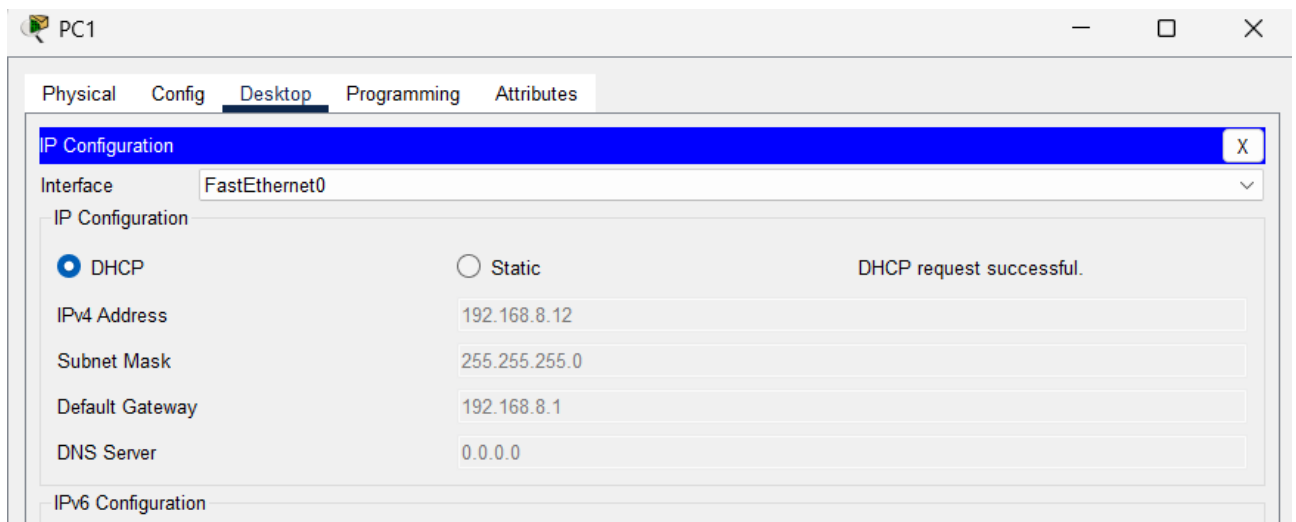
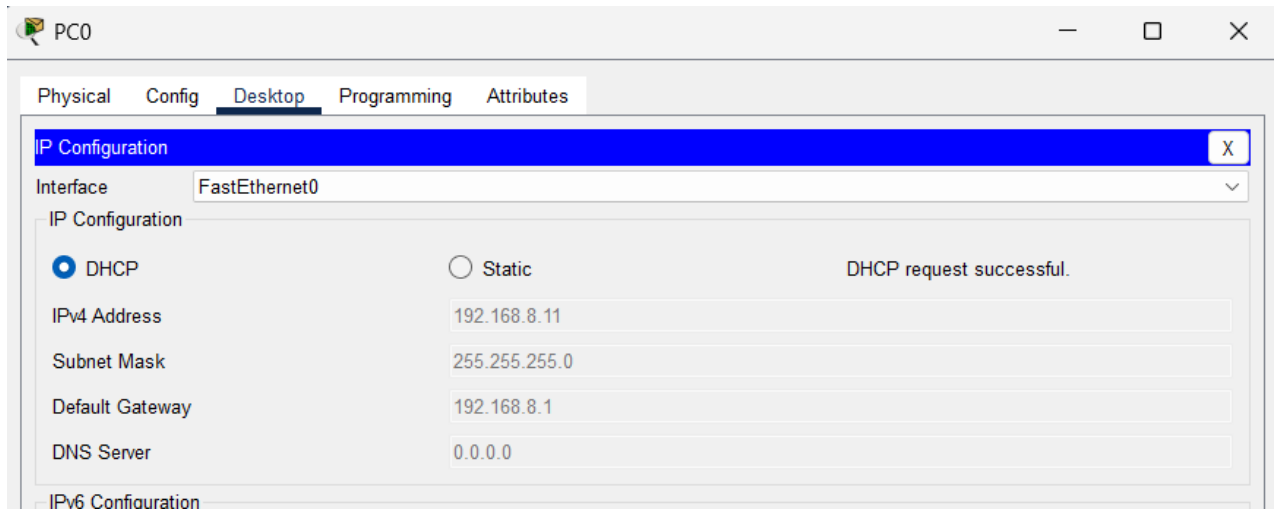
DHCP-Konfiguration auf dem Switch & IP-Routing aktivieren

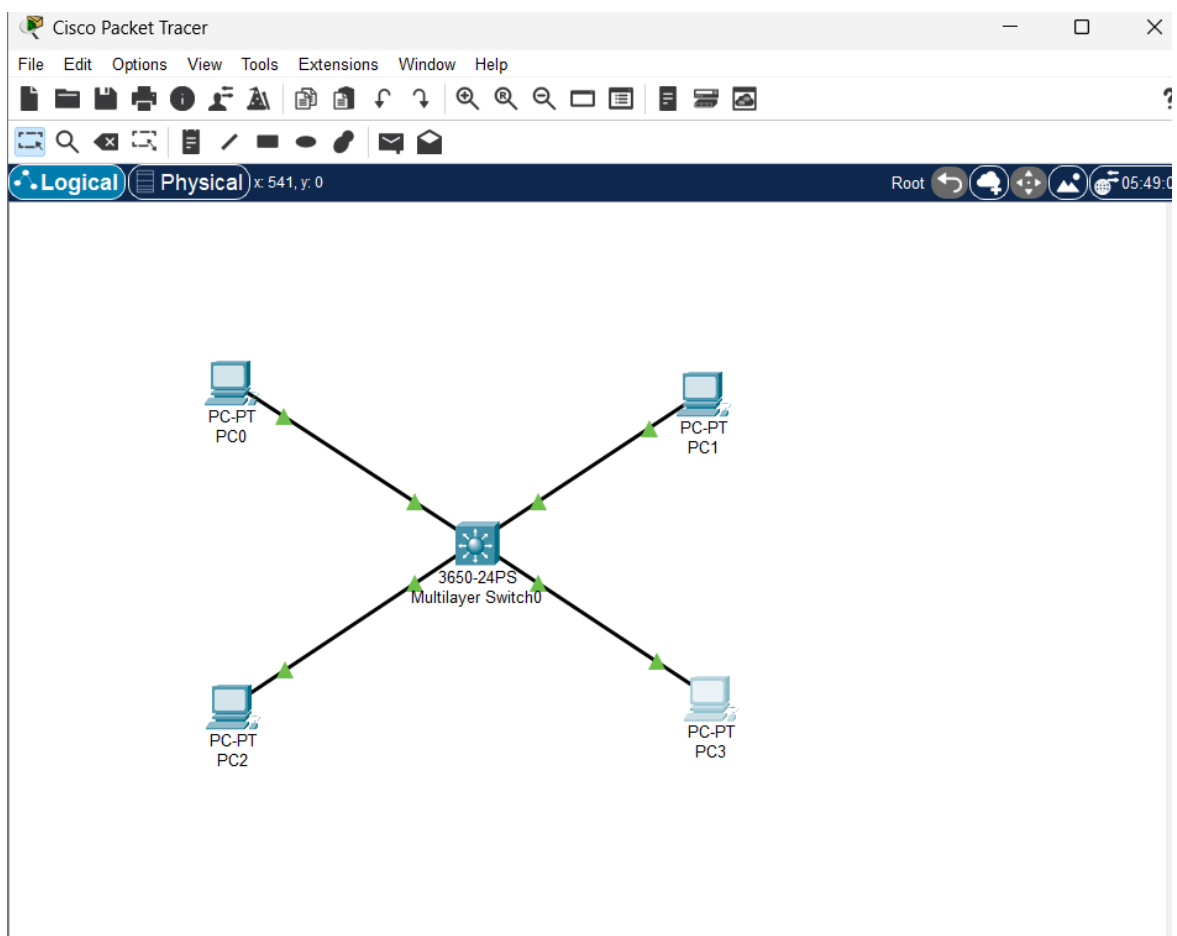
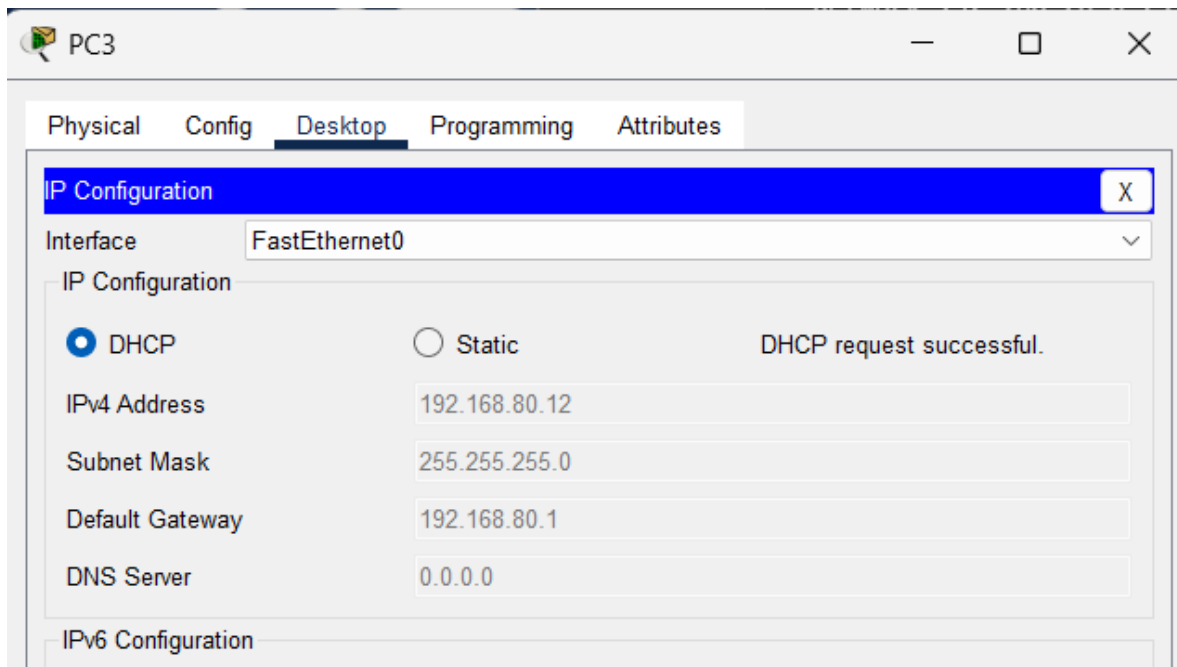
```
Switch(config)#
Switch(config)#ip routing
Switch(config)#
Switch(config)#ip dhcp pool VLAN10
Switch(dhcp-config)# network 192.168.8.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.8.1
Switch(dhcp-config)# exit
Switch(config)#
```

```
Switch(config)#
Switch(config)#ip dhcp pool VLAN20
Switch(dhcp-config)# network 192.168.80.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.80.1
Switch(dhcp-config)# exit
Switch(config)#
```

```
Switch(config)#
Switch(config)#ip dhcp excluded-address 192.168.8.1 192.168.8.10
Switch(config)#ip dhcp excluded-address 192.168.80.1 192.168.80.10
Switch(config)#
Switch(config)#
```

3) Test und Ergebnis





```
PC1

C:\>ping 192.168.80.12

Pinging 192.168.80.12 with 32 bytes of data:

Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127

Ping statistics for 192.168.80.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
PC2

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.8.12

Pinging 192.168.8.12 with 32 bytes of data:

Reply from 192.168.8.12: bytes=32 time<lms TTL=127
Reply from 192.168.8.12: bytes=32 time<lms TTL=127
Reply from 192.168.8.12: bytes=32 time<lms TTL=127
Reply from 192.168.8.12: bytes=32 time<lms TTL=127

Ping statistics for 192.168.8.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
PC0

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.80.12

Pinging 192.168.80.12 with 32 bytes of data:

Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127
Reply from 192.168.80.12: bytes=32 time<lms TTL=127

Ping statistics for 192.168.80.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```